Fleet Performance & Delivery Efficiency Dashboard

Power BI Module End Project | Logistics Analytics

This project analyzes trip-level logistics data to understand delivery punctuality, fuel efficiency, and estimated cost per kilometre. The supplied workbook contains 50 trip records and 7 vehicle master records.

1. Data preparation and modelling

Three missing values were found in Fuel_Consumed_L. They were replaced using mean imputation; the mean of the available fuel values was 91.88 litres. Vehicle attributes were joined to trips using Vehicle_ID for the supporting analysis. The original Vehicle_Master values were retained.

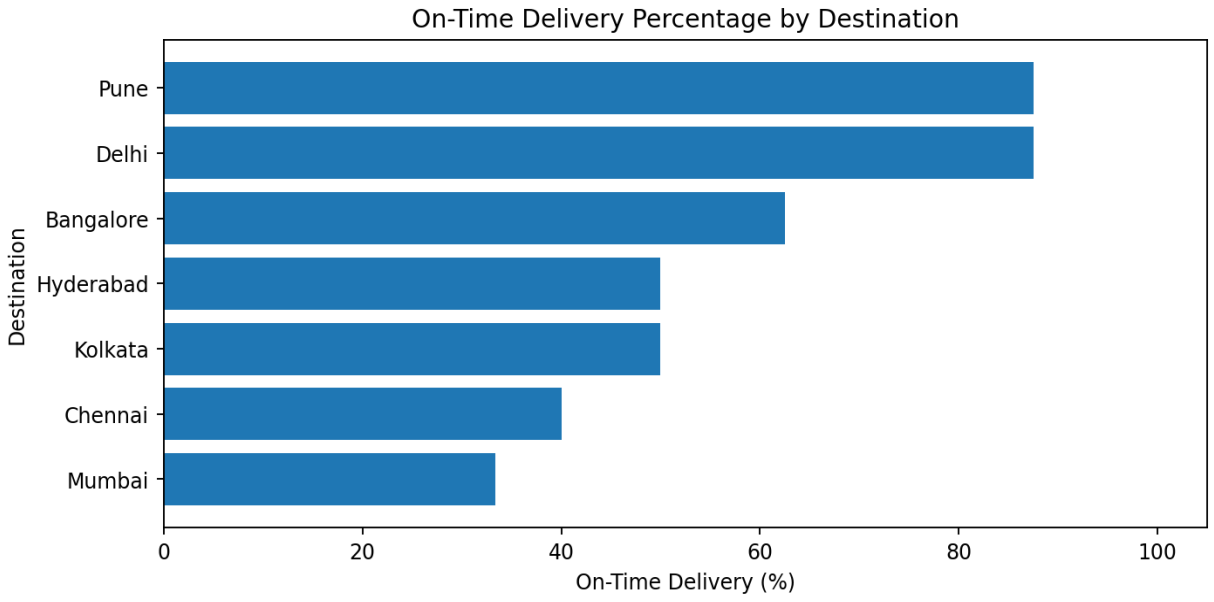
2. KPI summary

Metric	Calculated result
Total trips	50
On-time delivery rate	60.0%
Average fuel efficiency	12.07 km/L
Average cost per km	18.03 (formula units/km)
Average delivery time	Not available from supplied fields

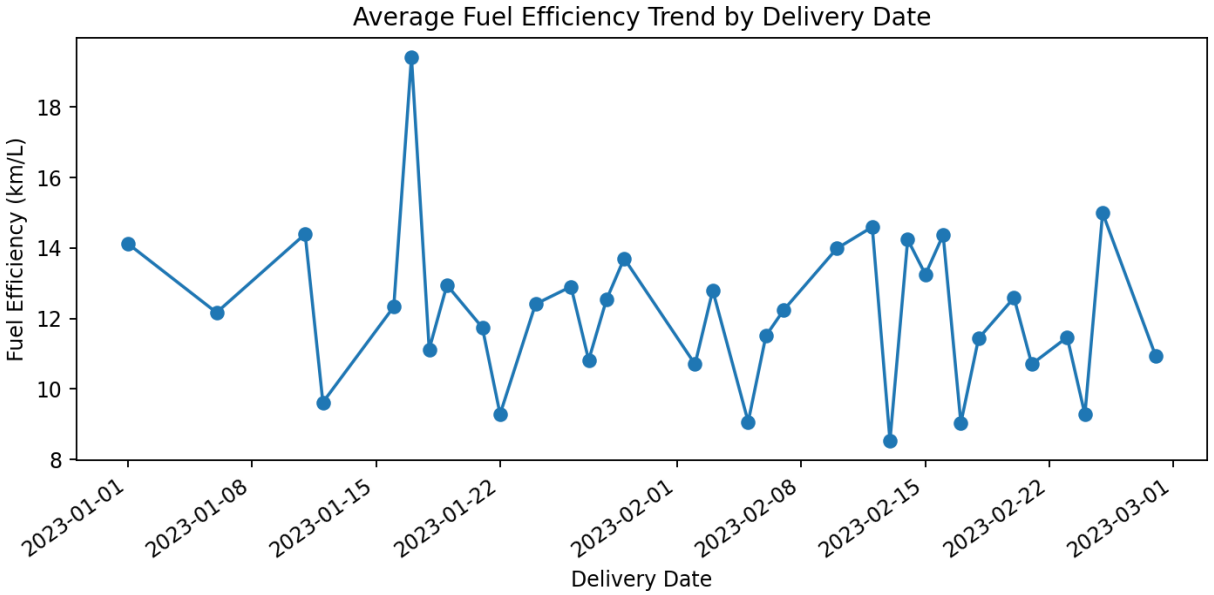
Cost interpretation: the assignment specifies Cost per km = (fuel cost + Maintenance Cost) / Distance, with fuel cost set to 100. This project follows that supplied formula; the result is an estimate based on the assignment's fixed fuel-cost assumption.

3. Visual analysis

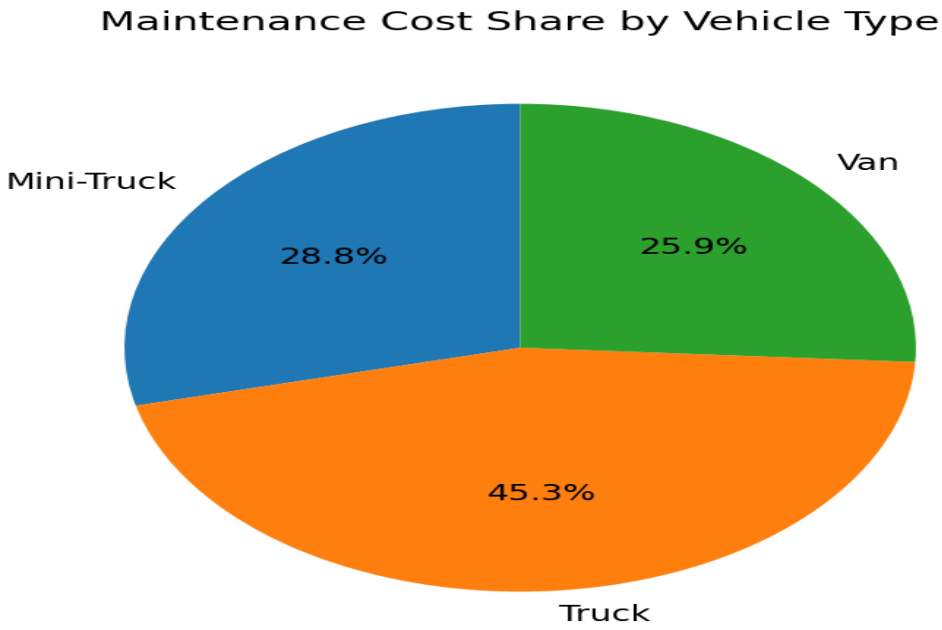
On-time delivery percentage by destination



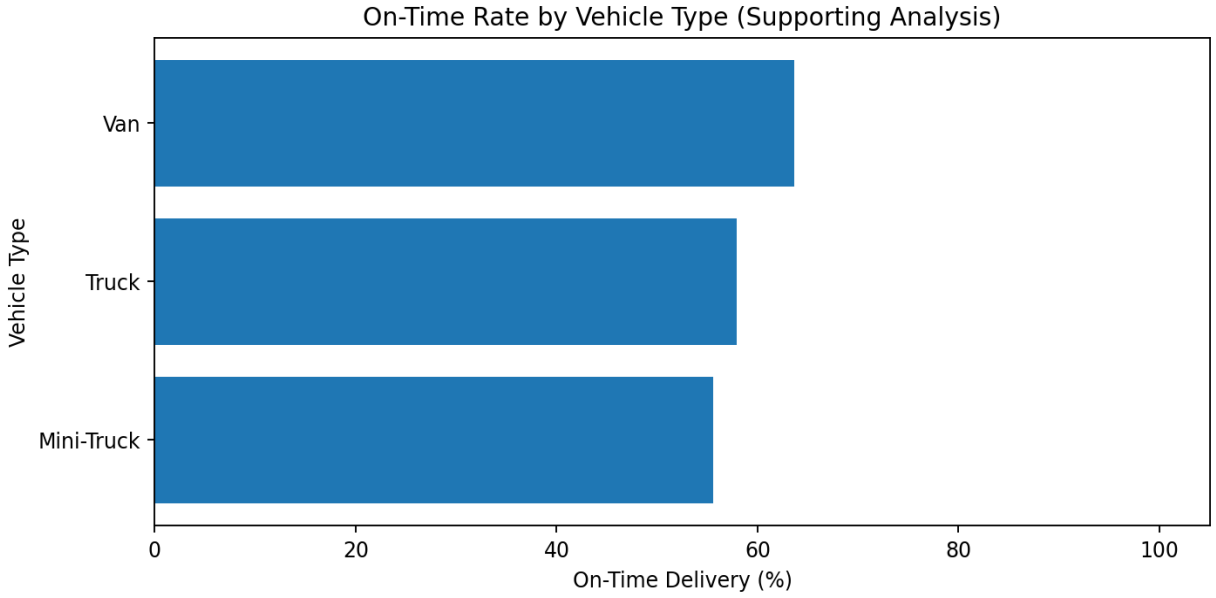
Fuel efficiency trend by delivery date


Average Fuel Efficiency Trend by Delivery Date
Fuel Efficiency (km/L)
18
16
14
12
10
8
2023-01-01
2023-01-08
2023-01-15
2023-01-22
2023-02-01
2023-02-08
2023-02-15
2023-02-22
2023-03-01
Delivery Date

Maintenance cost by vehicle type



Supporting analysis: on-time delivery rate by vehicle type



4. AI visual setup in Power BI

Visual	Fields / prompt
Q&A	Prompt: Average Cost per km by vehicle type?
Key Influencers	Analyze: Delivery_Status. Explain by: Distance_km, Vehicle_Type, Driver_ID.
Decomposition Tree	Analyze: Cost per km. Explain by: Vehicle_Type, Maintenance_Cost, Distance_km.

5. Data limitation and recommendation

The provided dataset has no delivery duration or delivery timestamp pair, so average delivery time cannot be derived reliably. For a complete KPI, request a delivery duration field or planned/actual delivery timestamps from the data owner. Also confirm the currency/unit for the fixed fuel cost of 100 before interpreting cost per km as a financial measure.

6. Power BI build checklist

• Load Trip_Data_Cleaned and Vehicle_Master.

• Create a one-to-many relationship on Vehicle_ID (Vehicle_Master on the one side).

• Create measures from DAX_Measures_and_PowerBI_Build_Notes.txt.

• Add bar chart, line chart, KPI cards, pie chart, Q&A;, Key Influencers, and Decomposition Tree as specified in the assignment.

• Use the calculated analysis workbook to validate figures and capture genuine screenshots from Power BI Desktop.